

Multivariate Analysis of Scotch Whisky Components

To Cover: - A high-level description of the study and its motivation. - A quantitative descriptive analysis of the data. - A graphical/qualitative analysis of the data. - Can you assume that the data follow a multivariate normal distribution? - no - Are there “surprising” observations? You must try to answer the question (with a proper discussion): Is the cluster analysis consistent with different distances and linkages?

Abstract

Introduction

High level description of the study and its motivation

Seeing if it's possible to classify whiskey as counterfeit using trace element analysis from total reflection x-ray fluorescence (TXRF). The study also attempted cold ashing, but causes contamination - and they deemed it is unnecessary

As an alternative to ICP How good was it - pretty good it seems, can correctly discriminate between non-counterfeit and counterfeit observations with a linear discriminator - TXRF should be easier to carry out compared to ICP

Techniques used in the study are: - PCA - dendrograms - linear discriminator

Data

The data contains contains 32 observations across: - Counterfeit - Blend - Grain - Highland - Lowland - Speyside

With trace elements for, (P, S, Cl, K, Ca, Mn, Fe, Cu, Zn, Br and Rb), measured in mg/L.

12 out of the 32 observations have at least one of the trace element detected below LOD. All the counterfeit whiskey observations also have at least one variables below LOD

Amount of observations below LOD by individual element:

P	S	Cl	K	Ca	Mn	Fe	Cu	Zn	Br	Rb
10	0	1	0	0	0	0	0	0	1	3

Transformations on Data

Handling the LOD values

If we were to drop observations with trace elements that were measured below LOD, we'd lose more than a third of the dataset, including all counterfeit observations, because of this they will instead be substituted.

The two common ways of substituting LOD values are, $LOD/2$ and $LOD/\sqrt{2}$. Similar to the paper that this data is from, this report will be substituting all LOD values with $LOD/2$.

S		Cl		K		Ca	
Min.	: 0.576	Min.	:0.0415	Min.	: 0.336	Min.	:0.4160
1st Qu.	: 1.080	1st Qu.	:0.1427	1st Qu.	: 3.110	1st Qu.	:0.7432
Median	: 2.585	Median	:0.1880	Median	: 6.165	Median	:1.0450
Mean	: 5.019	Mean	:0.2691	Mean	:10.262	Mean	:1.1918
3rd Qu.	: 5.540	3rd Qu.	:0.3432	3rd Qu.	:12.800	3rd Qu.	:1.4000
Max.	:26.100	Max.	:1.3500	Max.	:37.700	Max.	:4.1300
Mn		Fe		Cu		Zn	
Min.	:0.00600	Min.	:0.01900	Min.	:0.0380	Min.	:0.00700
1st Qu.	:0.01175	1st Qu.	:0.04925	1st Qu.	:0.1830	1st Qu.	:0.01875
Median	:0.01950	Median	:0.08800	Median	:0.2405	Median	:0.02300
Mean	:0.02322	Mean	:0.09456	Mean	:0.3799	Mean	:0.03716
3rd Qu.	:0.03125	3rd Qu.	:0.12900	3rd Qu.	:0.5255	3rd Qu.	:0.03725
Max.	:0.05300	Max.	:0.28800	Max.	:1.3200	Max.	:0.19400
Br		Rb					
Min.	:0.001000	Min.	:0.00050				
1st Qu.	:0.003000	1st Qu.	:0.00200				
Median	:0.004000	Median	:0.00600				
Mean	:0.007781	Mean	:0.00925				
3rd Qu.	:0.007000	3rd Qu.	:0.01325				
Max.	:0.068000	Max.	:0.03900				

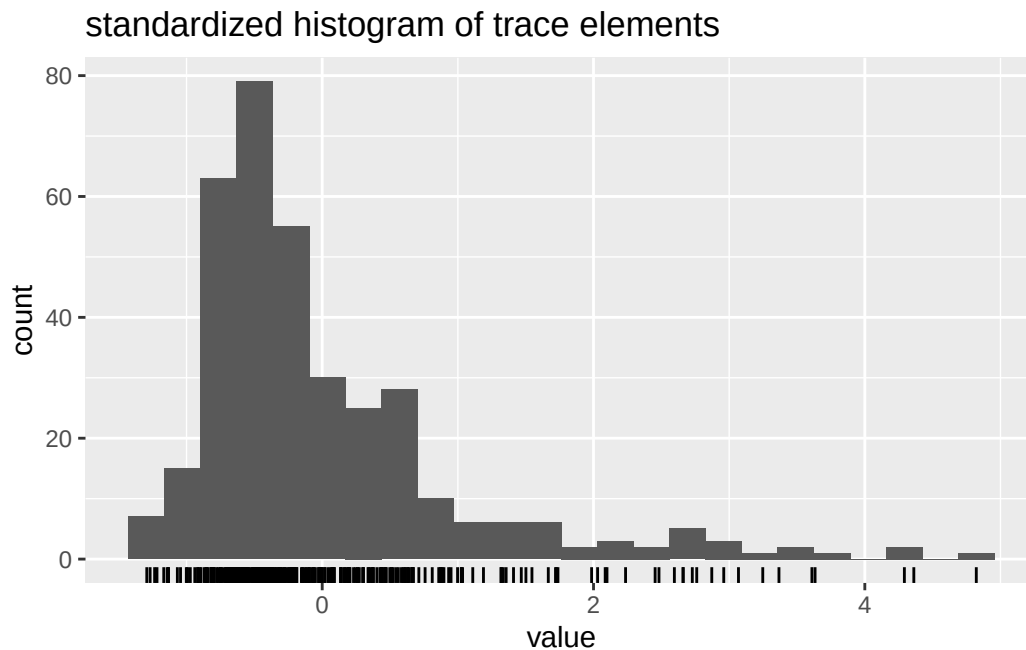
From the summary we can see all values are above 0, but they aren't all in the same magnituded of range. E.g., Br is a lot smaller than P.

Log Transform on the data

Lastly, looking at (standardized) histogram of all the variables, we can see it looks rather skew,

```
whiskey.stand <- whiskey.data |>
  mutate(across(where(is.numeric), ~(scale(.) %>% as.vector)))
whiskey.stand$Descriptor <- whiskey.data$Descriptor
whiskey.stand.melt <- melt(whiskey.stand[-2],
  id.vars="Descriptor",
  variable.name = "Element")

ggplot(whiskey.stand.melt, aes(x=value))+
  geom_histogram(bins=nclass.FD(whiskey.stand.melt$value))+
  geom_rug() +
  ggtitle("standardized histogram of trace elements")
```



After applying a log transform, the data looks a lot closer to a normal distribution.

```
whiskey.stand <- whiskey.data |>
  mutate(across(where(is.numeric), log)) |>
  mutate(across(where(is.numeric), ~(scale(.) %>% as.vector)))
whiskey.stand.melt <- melt(whiskey.stand[-2],
  id.vars="Descriptor",
```

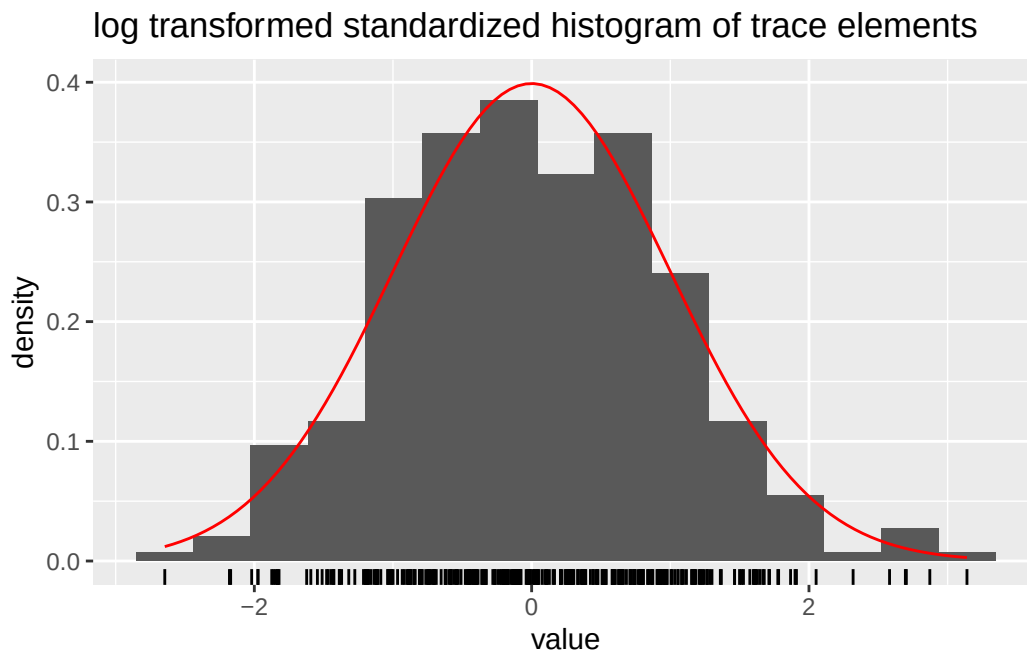
```

      variable.name = "Element")

ggplot(whiskey.stand.melt, aes(x=value))+
  geom_histogram(aes(y = ..density..), bins=nclass.FD(whiskey.stand.melt$value))+
  stat_function(fun = dnorm, args = list(mean=0, sd=1), color="red") +
  geom_rug() +
  ggtitle("log transformed standardized histogram of trace elements")

```

Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
 i Please use `after\_stat(density)` instead.



A Qualitative Analysis of Data

Box Plots

First looking at box plots of elements across Descriptor:

```

whiskey.melt <- melt(whiskey.data[,-2],
  id.vars="Descriptor",
  variable.name = "Element")
uniq.desc <- unique(whiskey.data$Descriptor)

```

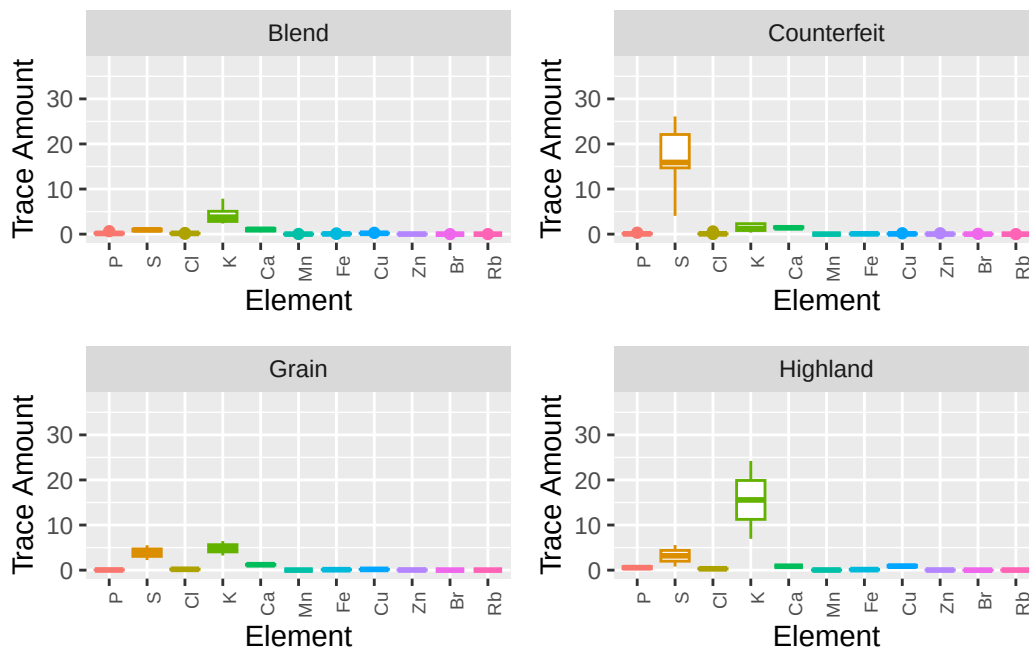
```

get.boxplots <- function(data, y_min, y_max){
  ggplot(data=data, aes(x=Element, y=value)) +
    geom_boxplot(aes(col=Element), notch = F) +
    facet_grid(~ Descriptor) +
    ylab("Trace Amount") +
    ylim(y_min, y_max) +
    theme(axis.text.x = element_text(size=7, angle=90, hjust=1),
          legend.position = "none")
}

plots <- lapply(1:7, function(i) {
  get.boxplots(subset(whiskey.melt, Descriptor == uniq.desc[i]), 0, max(whiskey.melt$value))
})

grid.arrange(grobs = plots[1:4], nrow = 2, main="log transformed Trace Element based on Descr

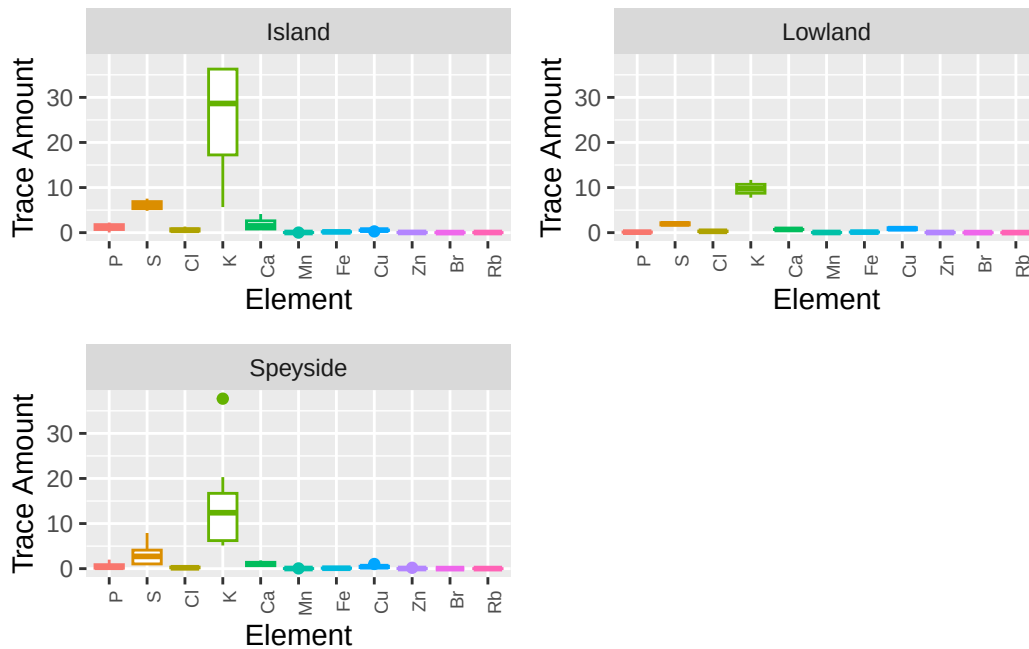
```



```

grid.arrange(grobs = plots[5:7], nrow = 2, main="log transformed Trace Element based on Descr

```

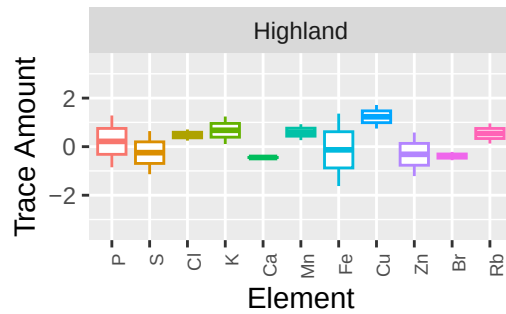
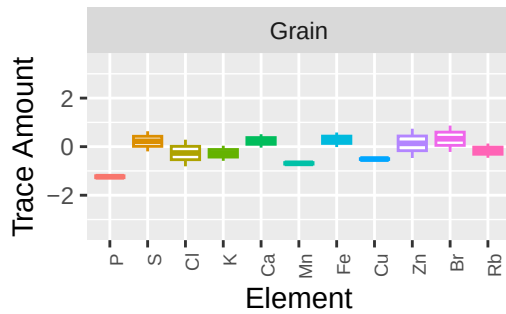
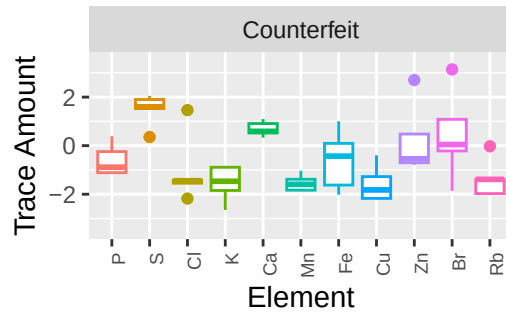
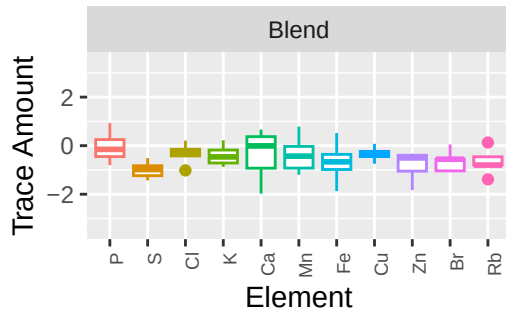


Because K's magnitude is so great compared to the other trace elements, standardizing the data helps better compare across all trace elements.

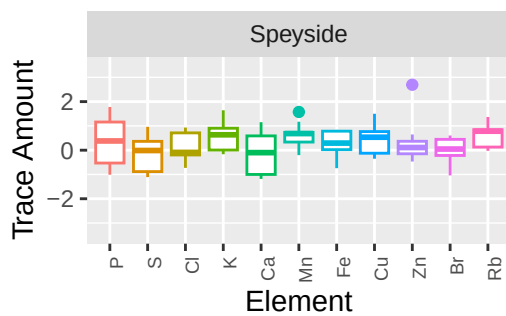
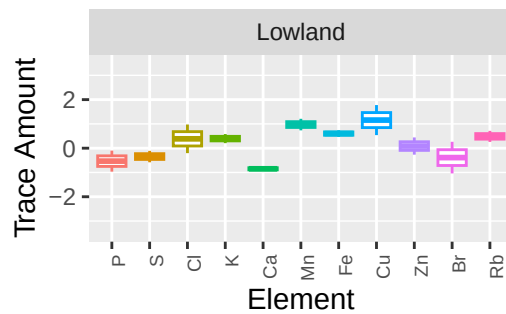
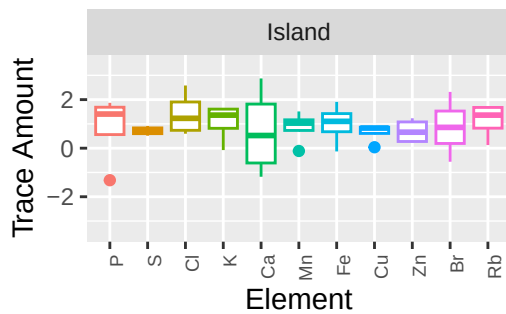
```
whiskey.stand.melt <- melt(whiskey.stand[,-2],
  id.vars="Descriptor",
  variable.name = "Element")

plots <- lapply(1:7, function(i) {
  get.boxplots(subset(whiskey.stand.melt, Descriptor == uniq.desc[i]), -3.5, 3.5)
})

grid.arrange(grobs = plots[1:4], nrow = 2, main="log transformed and standardized Trace Elements")
```



```
grid.arrange(grobs = plots[5:7], nrow = 2, main="log transformed and stranderdized Trace Ele
```



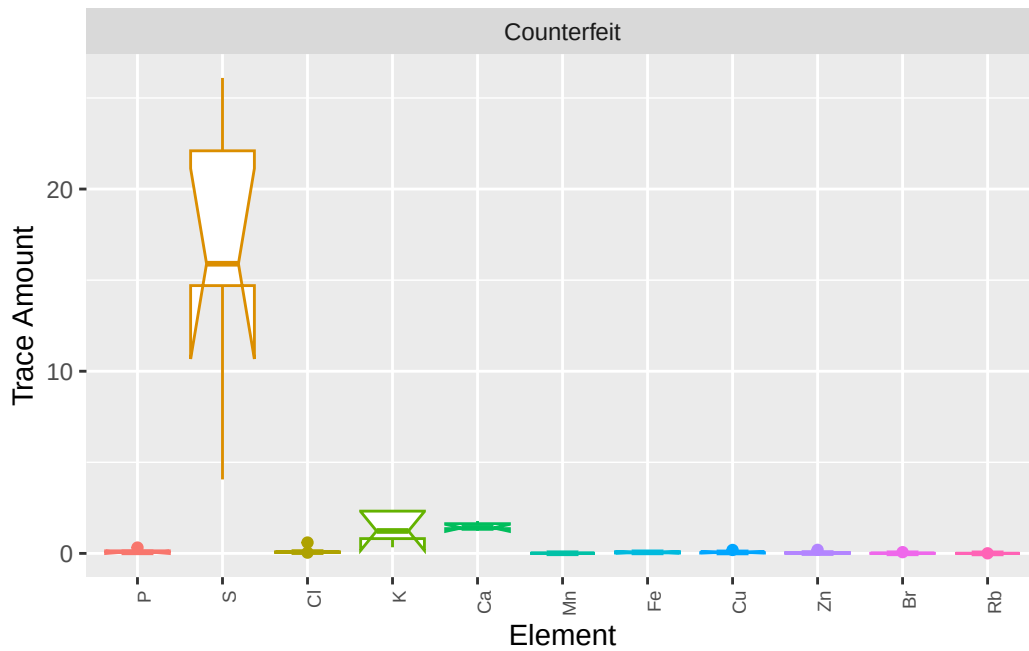
Looking at the boxplots there seem to be some differences. Counterfeit stands out a lot because of the high amount of S it compared to the others. Lowland has a suprisingly little amount of variance across all of it's elements.

Note on Notches for box plots

As the sample size is tiny, any attempt at adding notches causes it to go outside the range of the hinges (outside of the IQR). For an example:

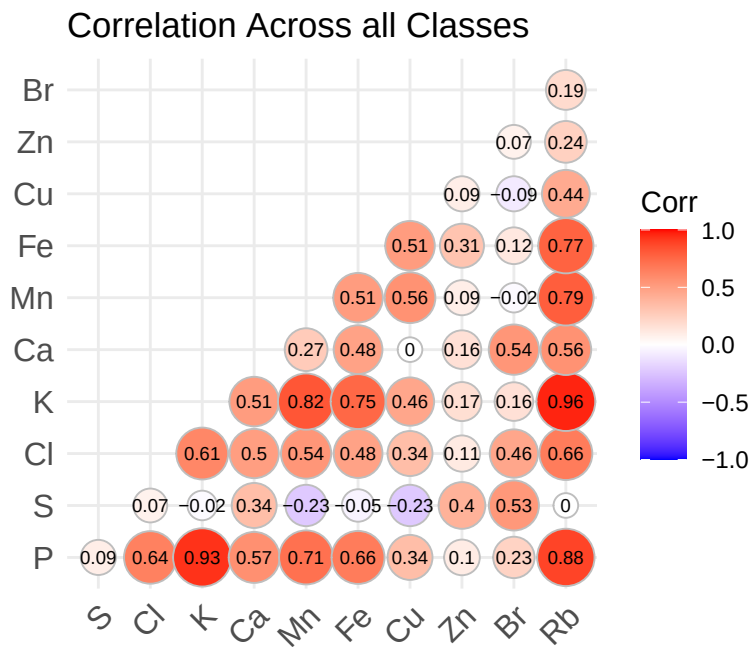
```
ggplot(data=subset(whiskey.melt, Descriptor=="Counterfeit"), aes(x=Element, y=value)) +  
  geom_boxplot(aes(col=Element), notch = T) +  
  facet_grid(~ Descriptor) +  
  ylab("Trace Amount") +  
  theme(axis.text.x = element_text(size=7, angle=90, hjust=1),  
        legend.position = "none")
```

```
Notch went outside hinges  
i Do you want `notch = FALSE`?  
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Notch went outside hinges  
i Do you want `notch = FALSE`?
```

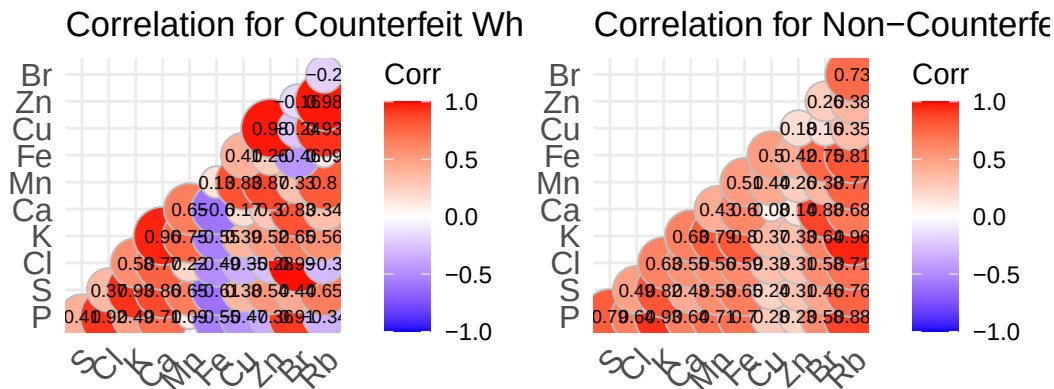
Correlograms

```
get.ggcorrplot(cor(whiskey.data[,-(1:2)]), "Correlation Across all Classes")
```



K, **P**, and **RB** all have strong positive correlations with one another,

```
counterfit.corrplot <- get.ggcorrplot(cor(subset(whiskey.data, Descriptor=="Counterfeit"),-
non.counterfit.corrplot <- get.ggcorrplot(cor(subset(whiskey.data, Descriptor!="Counterfeit"),-
grid.arrange(counterfit.corrplot, non.counterfit.corrplot, ncol=2)
```



There's a clear indicator for some difference with a lot more negative correlations for the counterfeit whiskeys compared to the others. Particularly along **Fe**. Although, the strongest correlations are still those positive ones.

When separating out based on Counterfeit vs Non-Counterfeit, Non-Counterfeit still maintains those strong correlations, with the greatest still being that between, **K**, **P**, and **RB**. This might be a common trace element picked up during whiskey production.

Due to the unbalanced observations, it makes sense that the original correlogram is similar to the non-counterfeit correlogram.

It's worthwhile also checking across more descriptor labels

Mahalanobis:

```
cov(whiskey.num)
```

	S	Cl	K	Ca	Mn
S	39.2043185151	0.114812054	-1.44632861	1.4680571371	-1.952918e-02
Cl	0.1148120539	0.061428066	1.60478431	0.0840056331	1.776515e-03
K	-1.4463286099	1.604784306	111.70165457	3.6702059758	1.15539e-01
Ca	1.4680571371	0.084005633	3.67020598	0.4687516774	2.449411e-03

Mn	-0.0195291845	0.001776515	0.11555388	0.0024494113	1.773377e-04
Fe	-0.0193017278	0.007048293	0.47217644	0.0195556290	4.011956e-04
Cu	-0.4792100927	0.027898010	1.59319734	-0.0008030968	2.422996e-03
Zn	0.1086028821	0.001203004	0.07569622	0.0047587823	5.254536e-05
Br	0.0418237974	0.001427163	0.02119748	0.0046221048	-3.789315e-06
Rb	-0.0001482016	0.001655335	0.10232441	0.0038788710	1.058952e-04
	Fe	Cu	Zn	Br	Rb
S	-0.0193017278	-0.4792100927	1.086029e-01	4.182380e-02	-1.482016e-04
Cl	0.0070482933	0.0278980101	1.203004e-03	1.427163e-03	1.655335e-03
K	0.4721764415	1.5931973427	7.569622e-02	2.119748e-02	1.023244e-01
Ca	0.0195556290	-0.0008030968	4.758782e-03	4.622105e-03	3.878871e-03
Mn	0.0004011956	0.0024229960	5.254536e-05	-3.789315e-06	1.058952e-04
Fe	0.0035474153	0.0100396210	7.857802e-04	9.044960e-05	4.605806e-04
Cu	0.0100396210	0.1071605645	1.284472e-03	-3.645766e-04	1.455597e-03
Zn	0.0007857802	0.0012844718	1.863620e-03	3.858367e-05	1.045565e-04
Br	0.0000904496	-0.0003645766	3.858367e-05	1.559829e-04	2.329839e-05
Rb	0.0004605806	0.0014555968	1.045565e-04	2.329839e-05	1.010161e-04

```

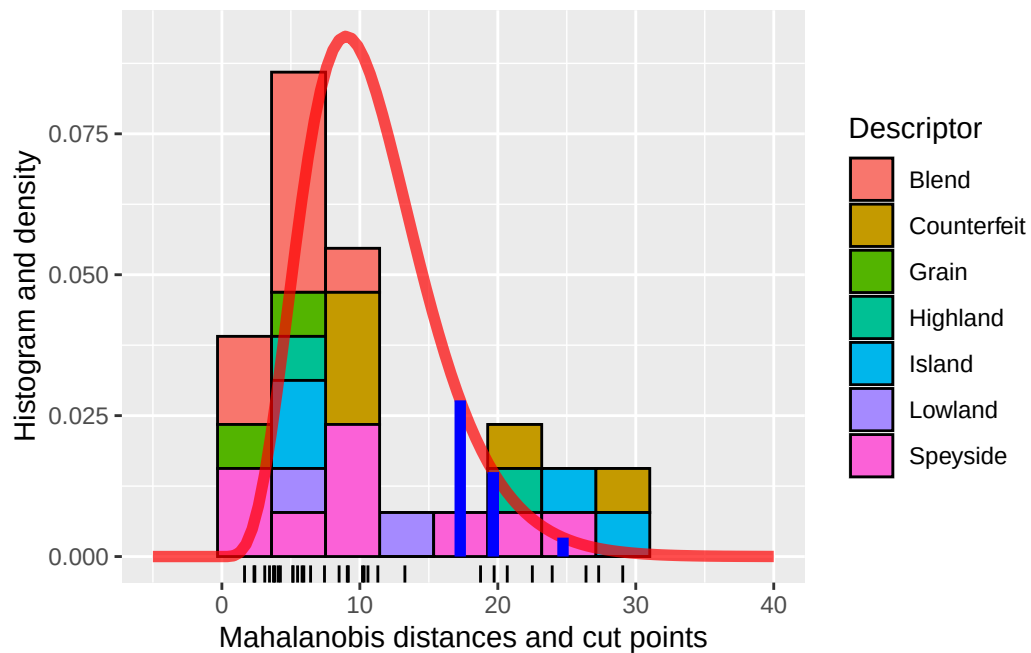
p <- 11
dM <- mahalanobis(whiskey.data[-(1:2)], center=colMeans(whiskey.data[-(1:2)]), cov=cov(whiskey.data[-(1:2)]))

upper.quantiles <- qchisq(c(.9, .95, .99), df=p)
density.at.quantiles <- dchisq(x=upper.quantiles, df=p)
cut.points <- data.frame(upper.quantiles, density.at.quantiles)

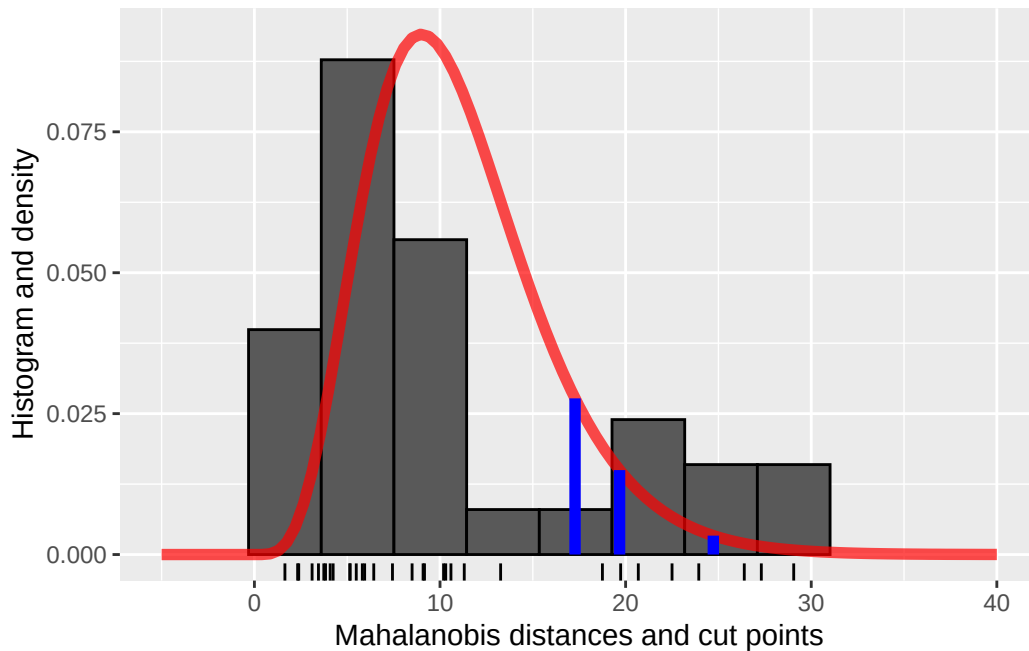
dM <- data.frame(dM)
dM$Descriptor <- whiskey.data$Descriptor
ggplot(dM, aes(x=dM)) +
  #geom_histogram(aes(y=after_stat(density)), bins=8, col="black") +
  geom_histogram(aes(fill=Descriptor, y=after_stat((count/sum(count))/4)), bins=8, col="black") +
  geom_rug() +
  stat_function(fun=dchisq, args = list(df=p),
    col="red", size=2, alpha=.7, xlim=c(-5,40)) +
  geom_segment(data=cut.points,
    aes(x=upper.quantiles, xend=upper.quantiles,
      y=rep(0,3), yend=density.at.quantiles),
    col="blue", size=2) +
  xlab("Mahalanobis distances and cut points") +
  ylab("Histogram and density")

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



```
ggplot(dM, aes(x=dM)) +
  geom_histogram(aes(y=after_stat(density)), bins=8, col="black") +
  #geom_histogram(aes(fill=Descriptor, y=after_stat(density/8)), bins=8, col="black") +
  geom_rug() +
  stat_function(fun=dchisq, args = list(df=p),
               col="red", size=2, alpha=.7, xlim=c(-5,40)) +
  geom_segment(data=cut.points,
              aes(x=upper.quantiles, xend=upper.quantiles,
                  y=rep(0,3), yend=density.at.quantiles),
              col="blue", size=2) +
  xlab("Mahalanobis distances and cut points") +
  ylab("Histogram and density")
```



The data seems to be forming 2 groups, with a dip in the middle (around 15 on the x axis). The only thing that stands out (for what might be the reason) is observations with 'Blend' as their descriptor are all to the left, dragging everything else with them.

Its important to note the histograms with descriptor colours isn't scaled 100%, ngl, I have no idea how to do it correctly, but for a qualitative analysis it's close enough.

TODO: see what those points on the far right are ### PCA

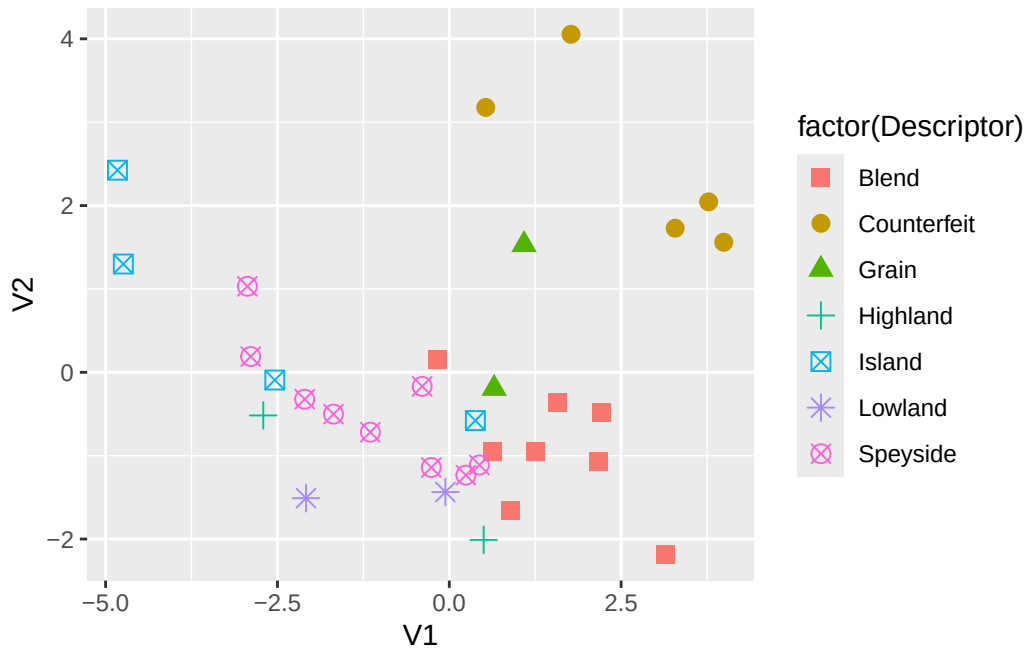
```
whiskey.log <- whiskey.data |> mutate(across(where(is.numeric), log))
whiskey.stand <- whiskey.log |> mutate(across(where(is.numeric), ~(scale(.) %>% as.vector)))

Sigma <- cov(whiskey.stand[-(1:2)])
#eigen(Sigma)
pca <- as.matrix(whiskey.stand[3:13]) %*% eigen(Sigma)$vectors

pca <- as.data.frame(pca)
pca$Descriptor <- whiskey.data$Descriptor

ggplot(pca, aes(x=V1,y=V2, color=factor(Descriptor), shape=factor(Descriptor)))+
  geom_point(size=3)+
  scale_shape_manual(values = c("Blend"=15,
                                "Counterfeit"=16,
                                "Grain"=17,
                                "Highland"=3,
```

```
"Island"=7,  
"Lowland"=8,  
"Speyside"=13))
```



There is a clear separation of the counterfeits from the rest in the upper right corner. Grain seems to be seeping into that cluster a bit tho.

```
pca2 <- prcomp(whiskey.stand[-(1:2)], center=TRUE, scale=TRUE)  
summary(pca2)
```

Importance of components:

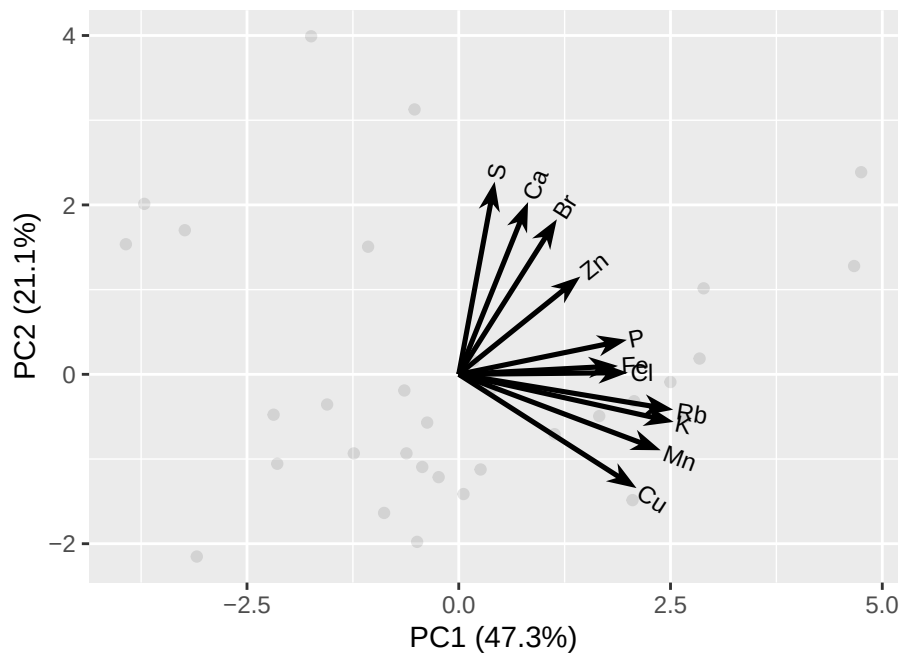
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.2801	1.5242	1.0951	0.86707	0.66526	0.60480	0.49676
Proportion of Variance	0.4726	0.2112	0.1090	0.06835	0.04023	0.03325	0.02243
Cumulative Proportion	0.4726	0.6838	0.7928	0.86119	0.90142	0.93468	0.95711

	PC8	PC9	PC10	PC11
Standard deviation	0.4337	0.41989	0.30011	0.13164
Proportion of Variance	0.0171	0.01603	0.00819	0.00158
Cumulative Proportion	0.9742	0.99024	0.99842	1.00000

```
pca2$rotation
```

	PC1	PC2	PC3	PC4	PC5	PC6
P	0.31924612	0.097902279	-0.2120991	0.548308225	-0.4074035	-0.25314862
S	0.06821158	0.547348073	0.2311134	0.210395591	0.1663372	-0.55567168
Cl	0.32116572	0.004685254	-0.4160660	-0.448679450	-0.0934019	-0.16342892
K	0.40817466	-0.135223577	-0.1112254	0.108624481	0.1853779	-0.19881070
Ca	0.13165642	0.489406943	-0.2908339	0.276487520	0.1177837	0.69073588
Mn	0.38346175	-0.215726408	-0.1068212	0.075797703	0.2868130	0.08295689
Fe	0.30232008	0.023820062	0.4770248	-0.001570168	-0.6435641	0.23811392
Cu	0.33758824	-0.323300917	0.1261756	-0.121665818	-0.1012310	0.04957821
Zn	0.23091370	0.277915821	0.5483864	-0.262856115	0.2841972	0.09431934
Br	0.18633290	0.440497018	-0.2534662	-0.514524933	-0.1860483	-0.05855964
Rb	0.40730144	-0.100474472	0.1080633	0.105614098	0.3568282	0.07369217
	PC7	PC8	PC9	PC10	PC11	
P	-0.52869122	-0.083990891	0.01374212	0.052852472	0.152499874	
S	0.42569948	-0.064619956	-0.23955641	0.137134474	0.007039065	
Cl	0.17855867	-0.657997370	0.08626514	-0.056947504	0.106981736	
K	0.06810047	0.186195148	0.20370771	-0.405506709	-0.686882950	
Ca	0.12212179	-0.134296674	-0.18704658	-0.058824485	-0.140472616	
Mn	0.11459330	0.129053270	0.20840316	0.791746388	-0.006246692	
Fe	0.37278420	0.009221927	0.26356842	0.007676463	-0.032042842	
Cu	-0.02656161	0.032501876	-0.85355614	0.031125182	-0.082421925	
Zn	-0.55131742	-0.278623606	0.09711992	0.048835360	-0.128675339	
Br	-0.13914520	0.614347047	-0.01634251	0.043246938	0.072781350	
Rb	0.10822873	0.173940948	0.08568159	-0.418469216	0.666329348	

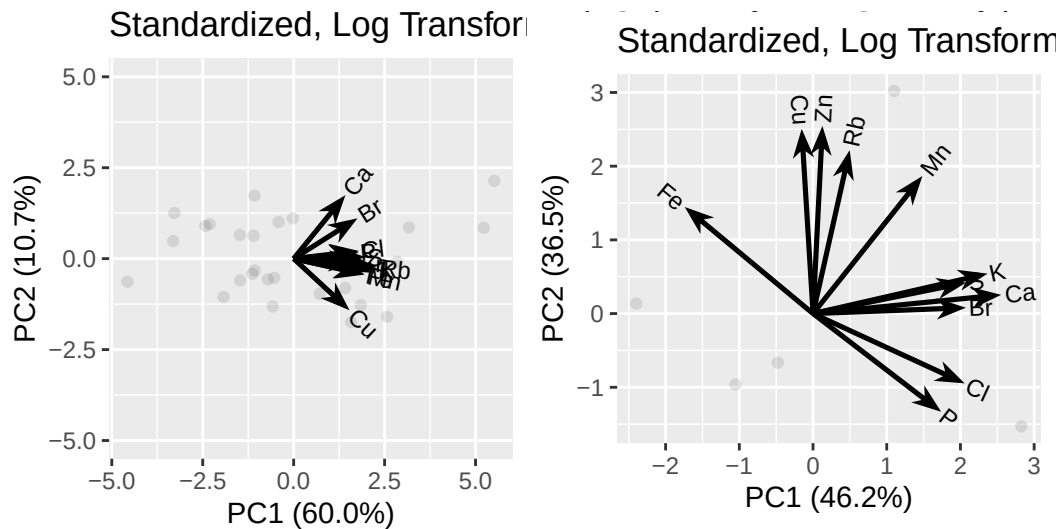
```
ggbiplot(pca2, obs.scale = 1, var.scale = 1, alpha = .1)
```



TODO: Interpret this more

```
pca2 <- prcomp(subset(whiskey.stand, Descriptor!="Counterfeit"), center=TRUE, scale=TRUE)
plot1 <- ggbiplot(pca2, obs.scale = 1, var.scale = 1, alpha = .1)+
  ylim(-5, 5)+
  ggtitle("Standardized, Log Transformed, Subset of Non-Counterfeit")

pca2 <- prcomp(subset(whiskey.stand, Descriptor=="Counterfeit"), center=TRUE, scale=TRUE)
plot2 <- ggbiplot(pca2, obs.scale = 1, var.scale = 1, alpha = .1)+
  ggtitle("Standardized, Log Transformed, Subset of Counterfeit")
#plot2
grid.arrange(plot1, plot2, ncol=2)
```

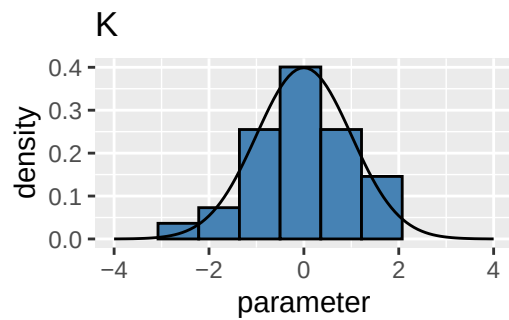
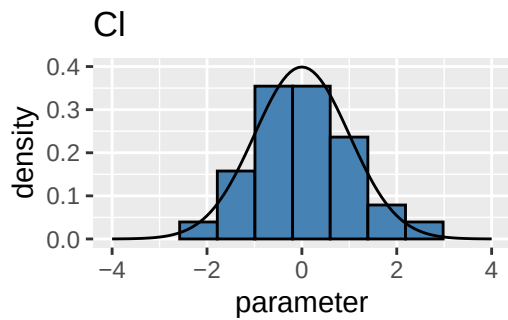
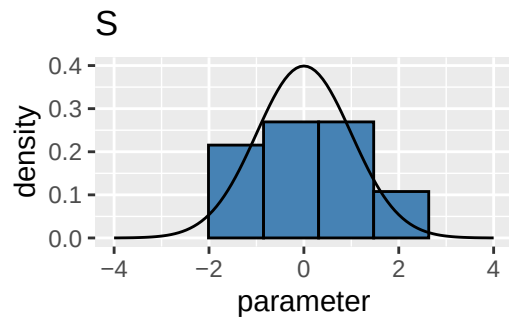
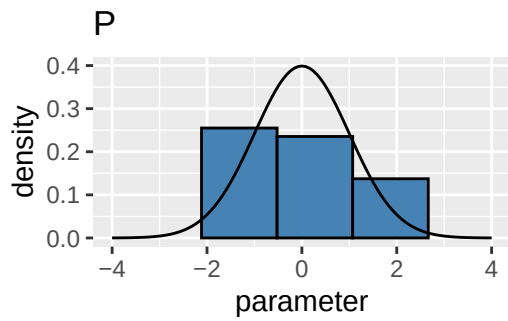
Once again, similar to the bar plots, immediately the **Fe** axis jumps out, being negative across PC1.

Multivariate Normality

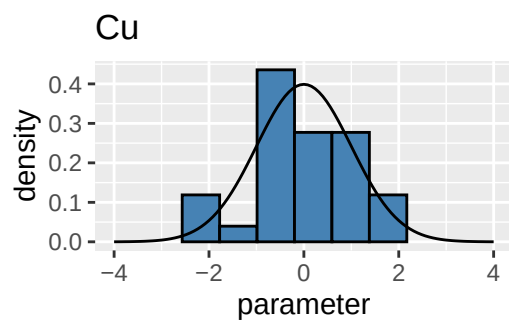
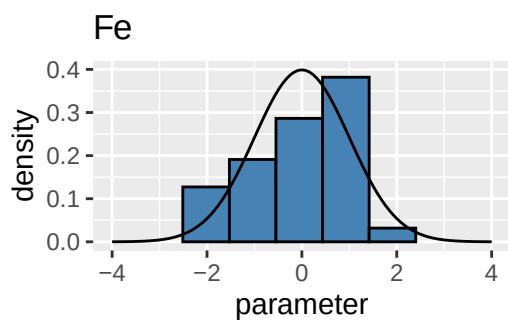
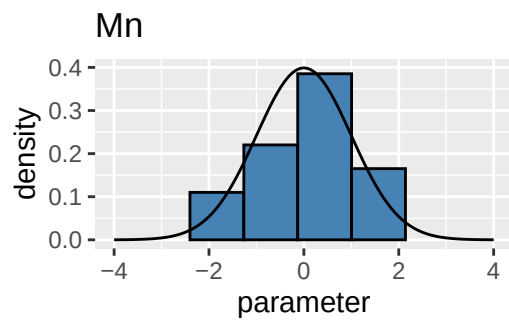
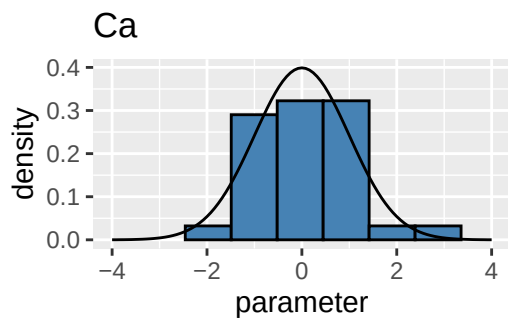
```
plot_norm_est <- function(data, parameter, t) {
  ggplot(data) +
    geom_histogram(mapping = aes(x = parameter, y = ..density..), fill = "steelblue", colour = "black") +
    stat_function(fun = dnorm, args = list(mean = 0, sd = 1), xlim = c(-4, 4)) +
    ggtitle(t)
}

plots <- list(plot_norm_est(whiskey.stand, whiskey.stand$P, "P"),
  plot_norm_est(whiskey.stand, whiskey.stand$S, "S"),
  plot_norm_est(whiskey.stand, whiskey.stand$Cl, "Cl"),
  plot_norm_est(whiskey.stand, whiskey.stand$K, "K"),
  plot_norm_est(whiskey.stand, whiskey.stand$Ca, "Ca"),
  plot_norm_est(whiskey.stand, whiskey.stand$Mn, "Mn"),
  plot_norm_est(whiskey.stand, whiskey.stand$Fe, "Fe"),
  plot_norm_est(whiskey.stand, whiskey.stand$Cu, "Cu"),
  plot_norm_est(whiskey.stand, whiskey.stand$Zn, "Zn"),
  plot_norm_est(whiskey.stand, whiskey.stand$Br, "Br"),
  plot_norm_est(whiskey.stand, whiskey.stand$Rb, "Rb"))

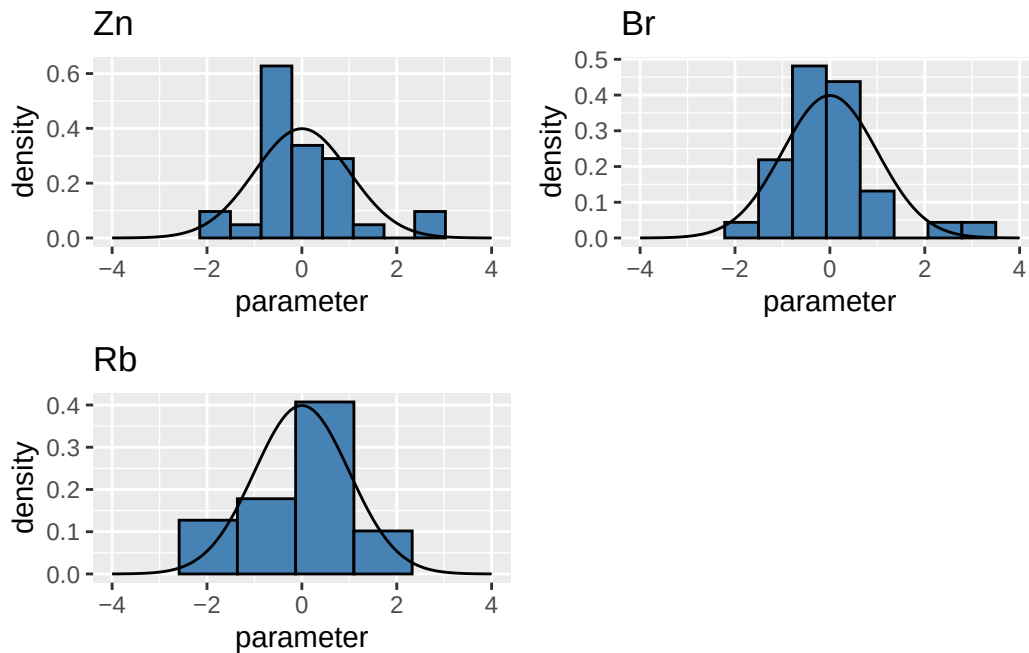
grid.arrange(grobs = plots[1:4], nrow = 2)
```



```
grid.arrange(grobs=plots[5:8], nrow=2)
```



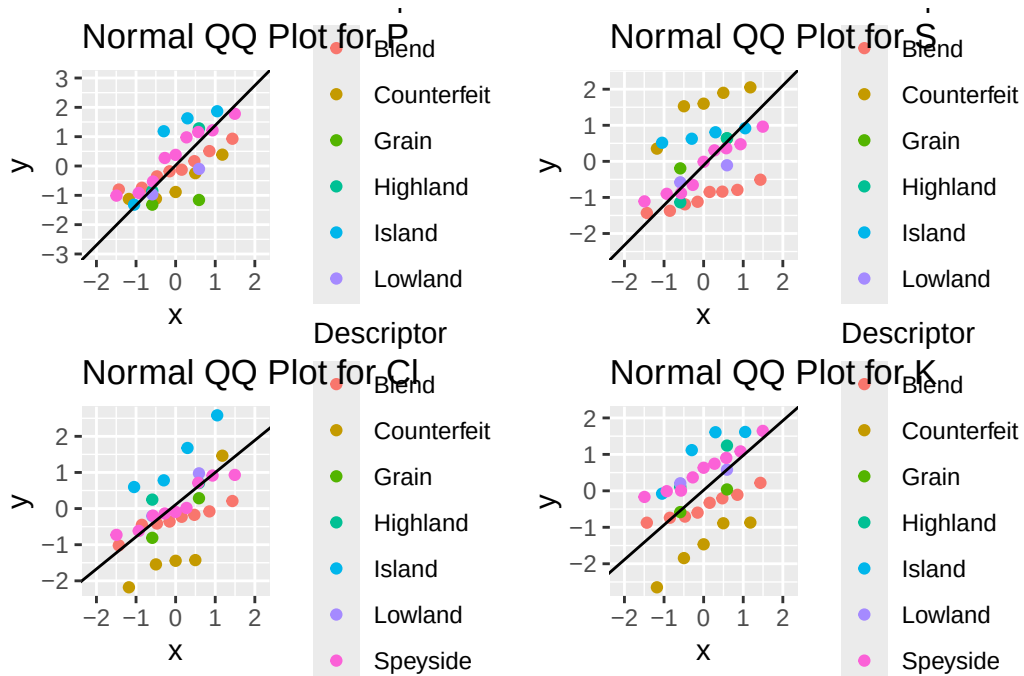
```
grid.arrange(grobs=plots[9:11], nrow=2)
```



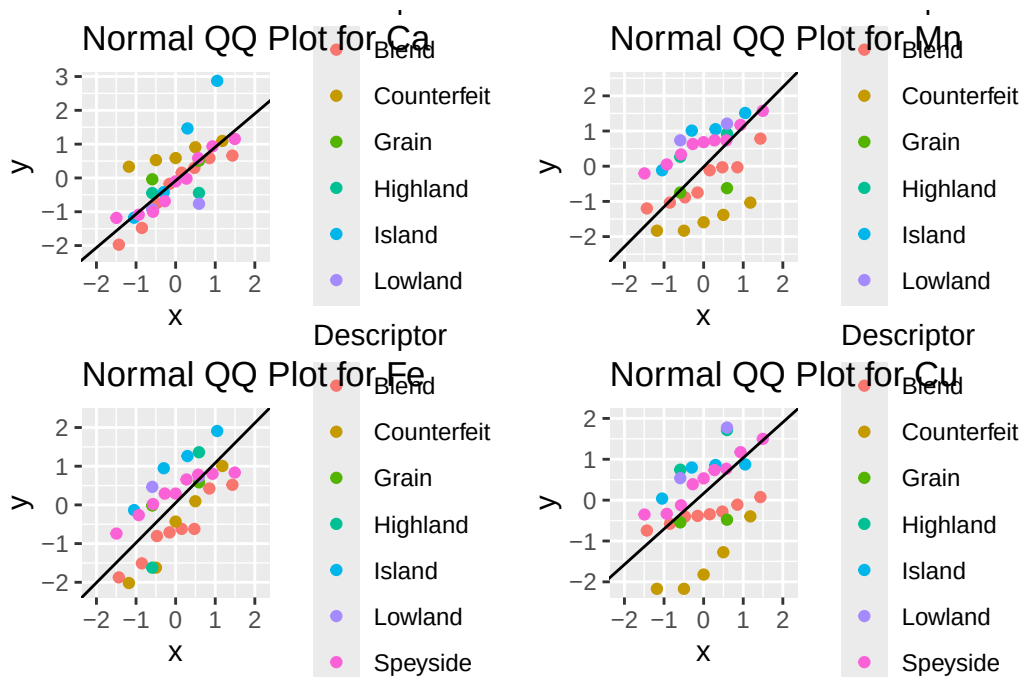
```
plot.qq <- function(data, parameter, title) {
  ggplot(data, aes(sample=parameter)) +
    stat_qq(mapping = aes(colour = Descriptor)) + stat_qq_line()+
    ggtitle(paste("Normal QQ Plot for", title, sep=" "))
}

plots <- list(plot.qq(whiskey.stand, whiskey.stand$P, "P")
,plot.qq(whiskey.stand, whiskey.stand$S, "S")
,plot.qq(whiskey.stand, whiskey.stand$Cl, "Cl")
,plot.qq(whiskey.stand, whiskey.stand$K, "K")
,plot.qq(whiskey.stand, whiskey.stand$Ca, "Ca")
,plot.qq(whiskey.stand, whiskey.stand$Mn, "Mn")
,plot.qq(whiskey.stand, whiskey.stand$Fe, "Fe")
,plot.qq(whiskey.stand, whiskey.stand$Cu, "Cu")
,plot.qq(whiskey.stand, whiskey.stand$Rb, "Rb")
,plot.qq(whiskey.stand, whiskey.stand$Zn, "Zn")
,plot.qq(whiskey.stand, whiskey.stand$Br, "Br"))

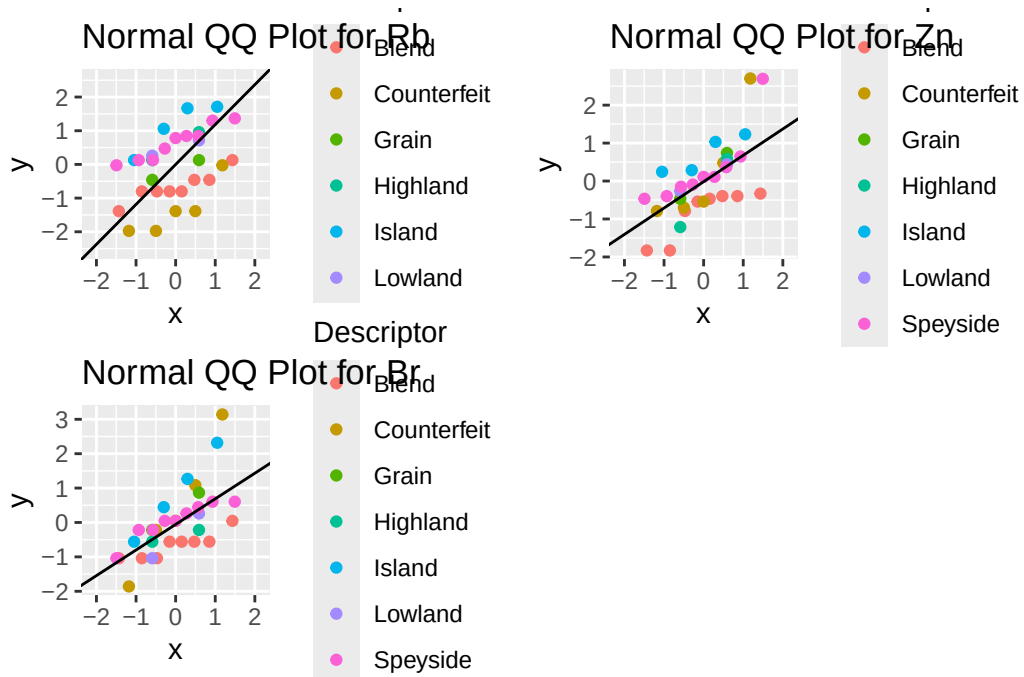
grid.arrange(grobs=plots[1:4], nrow=2)
```



```
grid.arrange(grobs=plots[5:8], nrow=2)
```



```
grid.arrange(grobs=plots[9:11], nrow=2)
```



Interestingly you can make out the counterfeits seem to be the strange observations in **S**, and **Cu**.

Apart from that the Q-Q plots look a-okay

TODO: Anderson Darling Test - can't remember the exact name :p

Classifying by distance

```
classify.data <- function(p, drop_i) {
  results <- data.frame(matrix(ncol=4,nrow=nrow(pca)))
  colnames(results) <- c('sample', 'true_class','estimated_class','is_correct')
  for (i in (1:nrow(pca))) {
    if (drop_i) {
      pca.dropped <- pca[-c(i),]
    } else {
      pca.dropped <- pca
    }

    pca.counter <- pca.dropped |> filter(Descriptor == "Counterfeit")
  }
}
```

```

pca.no.counter <- pca.dropped |> filter(Descriptor != "Counterfeit")

X1 <- pca.counter[(1:p)]
X2 <- pca.no.counter[(1:p)]

dM.counter <- mahalanobis(pca[i,(1:p)], center=colMeans(X1), cov=cov(X1))
dM.non.counter <- mahalanobis(pca[i,(1:p)], center=colMeans(X2), cov=cov(X2))

counter <- pchisq(dM.counter, df=p, lower.tail=FALSE)
non.counter <- pchisq(dM.non.counter, df=p, lower.tail=FALSE)

results[i, ]$sample = i
if (pca[i,]$Descriptor == as.factor("Counterfeit")) {
  results[i, ]$true_class = "Counterfeit"
  if (counter > non.counter) {
    # correct classification
    results[i, ]$estimated_class = "Counterfeit"
    results[i, ]$is_correct = TRUE
  } else {
    # incorrect classification
    results[i, ]$estimated_class = "Non - Counterfeit"
    results[i, ]$is_correct = FALSE
  }
} else {
  results[i, ]$true_class = "Non - Counterfeit"
  if (non.counter > counter) {
    # correct classification
    results[i, ]$estimated_class = "Non - Counterfeit"
    results[i, ]$is_correct = TRUE
  } else {
    # incorrect classification
    results[i, ]$estimated_class = "Counterfeit"
    results[i, ]$is_correct = FALSE
  }
}
}
results$true_class <- as.factor(results$true_class)
results$estimated_class <- as.factor(results$estimated_class)
results
}

```

```
dropped.i <- classify.data(p=2, drop_i=TRUE)
no.dropped.i <- classify.data(p=2, drop_i=FALSE)
```

Building confusion matrix:

```
get.conf.mat <- function(data) {
  conf.mat <- confusionMatrix(data$true_class, data$estimated_class)$table

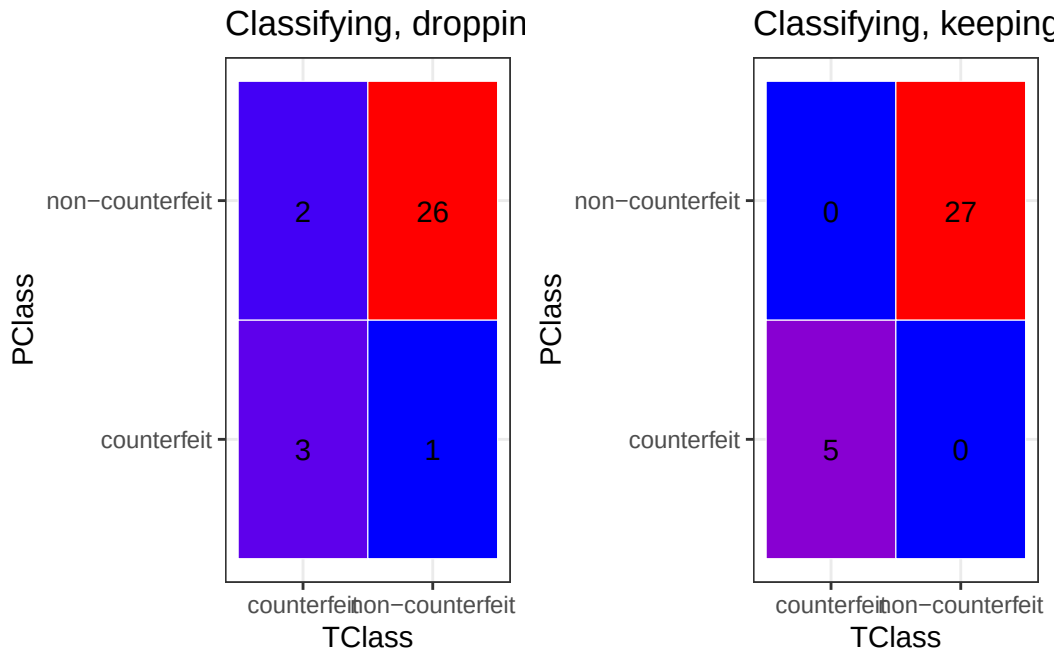
  ## From https://stackoverflow.com/questions/37897252/plot-confusion-matrix-in-r-using-ggplot2
  TClass <- factor(c("counterfeit", "counterfeit", "non-counterfeit", "non-counterfeit"))
  PClass <- factor(c("counterfeit", "non-counterfeit", "counterfeit", "non-counterfeit"))
  Y      <- c(conf.mat[1,1], conf.mat[1,2],
              conf.mat[2,1], conf.mat[2,2])
  df <- data.frame(TClass, PClass, Y)

  ggplot(data = df, mapping = aes(x = TClass, y = PClass)) +
    geom_tile(aes(fill = Y), colour = "white") +
    geom_text(aes(label = sprintf("%.0f", Y)), vjust = 1) +
    scale_fill_gradient(low = "blue", high = "red") +
    theme_bw() + theme(legend.position = "none")
}

plot1 <- get.conf.mat(dropped.i) +
  ggtitle("Classifying, dropping the ith row")

plot2 <- get.conf.mat(no.dropped.i) +
  ggtitle("Classifying, keeping the ith row")

grid.arrange(plot1, plot2, ncol=2)
```



using mahalanobis distance causes a fair bit of error when dropping the ith row for counterfeit observations. Might be because of the poor amount of data?

distance has to be done using PCA, since there would be too many factors compared to the amount of counterfeit observations ($p > 5$).

TODO: try classifying across all Descriptors